

High Altitude Balloon Software

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II. Abstract:

The Alpha CubeSat is a long-term project that ran out of Cornell's Space System Design Studio. The CubeSat, a 10x10x10 centimeter cubic satellite, aims to deploy a Solar Sail in orbit. The CubeSat has communication capabilities but can only downlink low-resolution images periodically. There is no way to capture video of the deployment in orbit. To get footage of the CubeSat ejecting a Solar Sail in a space-like environment, The Alpha Team started the high-altitude balloon project. A CubeSat will have all the sensory and power capabilities of the orbital CubeSat, but there will be no attitude control systems as the CubeSat will be launched in a launch rig equipped with cameras. This summer research aimed to get the launch system online for launch conditions in the fall. Successful completion of this project will allow the CubeSat team to get video footage of the light sail deploying in the upper atmosphere's extreme cold and vacuum-like environment.

Two goals of this project are the design and assembly of the mechanical, electrical, and software systems of the Balloon Rig. It is a continuation of the work previously started in the 2023-2024 school year. The flight board circuit board had undergone a new design iteration and had been fabricated. The flight board hosted a variety of sensory and communication systems that needed to be tested and programmed for the flight mission. The approach taken was to verify the electrical design functionality of the new board, as well as the code used to execute the mission. Finally, integration between the electrical and software systems with the mechanical assembly of the balloon rig began.

The backend development of all systems for the flight software has been completed. The sensory systems have all been validated. The GPS module can get a signal lock, and report accurate positional data. This normal report monitor adds all this data to a report, which is periodically sent down through the RockBLOCK control task. The camera control task successfully captures jpeg images using the serial camera and saves the files onboard the microcontroller SD card. The camera monitor loads these fragments from the SD card into a camera report, which is also transmitted. Finally, the commands sent from the RockBLOCK admin portal were received by the RockBLOCK modem and parsed into instructions through the control tasks.

Going forward, the next steps would be front-end work. The development of a ground station for the mission would be used to facilitate human interaction with flight software. It would be a user interface that allows for the transmission of commands, the display of normal report packages, and compiling and displaying the fragmented images.

III. Introduction

The Alpha CubeSat is a long-term project that ran out of Cornell's Space System Design Studio. The CubeSat, a 10x10x10 centimeter cubic satellite, aims to deploy a Solar Sail in orbit. The CubeSat has communication capabilities but can only downlink low-resolution images periodically. There is no way to capture video of the deployment in orbit. To get footage of the CubeSat ejecting a Solar Sail in a space-like environment, The Alpha Team started the high-altitude balloon project. A CubeSat will have all the sensory and power capabilities of the orbital CubeSat, but there will be no attitude control systems as the CubeSat will be launched in a launch rig equipped with cameras. This summer research aimed to get the launch system online for launch conditions in the fall. Successful completion of this project will allow the CubeSat team to get video footage of the light sail deploying in the upper atmosphere's extreme cold and vacuum-like environment.

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The launch attempt is scheduled for the upcoming fall semester.

IV. Materials and Methods

The materials used include a Balloon PCB previously designed in altium containing various passive elements, a gps module, radio communication modem, a burn wire, as well as temperature and light sensors. The board also contains usb breaks outs for external cameras used for filming. The board also features a simple serial camera capable of taking low resolution snapshots in orbit. First the electrical design of the PCB was verified by doing continuity checks. 5V is supplied via battery to the board. This voltage is regulated by the main system regulator, which supplies the microcontroller, radio modem, burn wire etc... Each USB breakout has its own 5V voltage regulator. These regulated voltage supplies were all verified before soldering components. All signal connections between components and the microcontroller were also confirmed before testing software. The components were secured using through-hole soldering.

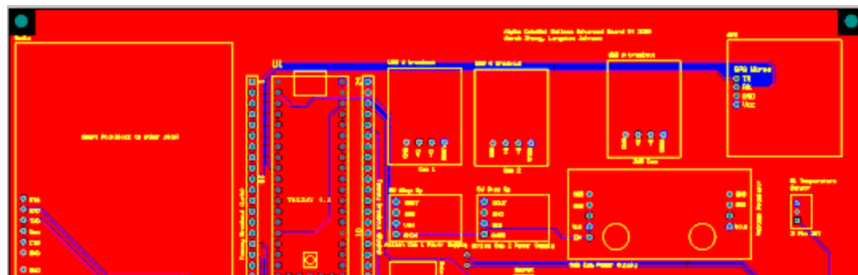


Figure 1: Balloon Flight Board V2

Figure 2: Flight board Soldering/Testing

The flight code is organized in the following structure:

- Each sensor signal is processed by a monitors
- The devices: radio modem, GPS module, and serial camera are operated using control tasks
- The main control loop executes all monitors and control tasks
- The mission manager transitions between mission states: standby, high altitude, deployment, and post-deployment. The current state determines the operations of the control tasks.

RockBLOCK and GPS Monitor:

The Radio Modem that interfaces the Iridium Satellite Constellation is called a RockBLOCK. RockBLOCK uses an AT command set. All commands start with AT followed by the specific command and end with a carriage return (\r). The RockBLOCK control task handles four main tasks using the RockBLOCK.

1. Comms Check
2. Binary Downlink of reports
3. Receive Response

In general, AT commands operate using a call-and-response structure. Each mode in the control task issues a command request and then handles the response data and result of each request.

The RockBLOCK modem sends Modem modem-originated buffers (MO) and receives Modem Terminated Buffers(MT).

The **Normal report** downlinks a variety of satellite-specific general data. They contain 70 bytes

Index	Data
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0	21(normal report flag)
1	altitude
2	longitude
3	latitude
4	Photoresistor covered
5	Mission mode
6	Burn Wire mode
7	Burn Wire attempts
8	Downlink period

*Table
1:*

Contents of a Normal Report

Camera Monitor and Camera Reports:

The Camera Monitor takes a snapshot of the environment using the Serial Camera. The image is broken down into fragments and saved to the SD card. The image is read from the SD file. The name of the image is “XXYY.jpeg” on the SD card. Where XX is the serial number and YY is the fragment number (leading zero format). When the camera takes a photo, it stores all the bytes in a buffer. Then it writes a fragment of length 64, or less and saves it to the SD card, and it saves how many fragments were necessary to write the image. Then the rockblock will downlink, fragment by fragment from the SD card to the ground station in a camera report.

Index	Data
0	42(camera report flag)
1	Serial Number
2-6 (4-bytes)	Fragment ID number
6-70(64-bytes)	Image Data

Table 2: Contents of a Camera Report

Commands:

There are three types of commands, each defined by an operation code(OP Code). Each OP code takes in one or two corresponding arguments. A command payload consists of 10 bytes. 2 opcode bytes, and 8 arg bytes(4 bytes per arg). Up to 4 commands can be sent at a time.

2-byte opcode	Byte 1	Byte 2
mission_mode	0x00	0x00
rockblock_downlink_period	0x01	0x00
request_image_fragment	0x02	0x00

Table 3: OPCode Binary

4-byte arg code	Byte 1	Byte 2	Byte 3	Byte 4
standby	0x00	0x00	0x00	0x00
high alititude	0x01	0x00	0x00	0x00
deployment	0x02	0x00	0x00	0x00
post_deployment	0x03	0x00	0x00	0x00
true_arg	0x01	0x00	0x00	0x00
false_arg	0x00	0x00	0x00	0x00
active	0x01	0x00	0x00	0x00
inactive	0x00	0x00	0x00	0x00
no_arg_2	0x00	0x00	0x00	0x00

Table 4: Argument Binary

OP Code	Arg 1	Arg 2
mission_mode	MISSION MODE	no_arg_2
rockblock_downlink_period	Downlink period(ms)	no_arg_2
request_image_fragment	Serial num	Frag num

Table 5: OPCODEs and Arguments

Results:

The temperature and light sensors were tested by running each corresponding monitor in the flight code to confirm the analog signals were converted into their proper values. These values were all added to a new normal report. The Burn Wire control tasks were executed to verify the nichrome wire could successfully burn through thin string fibers used in the release mechanism.

Confirming these functionalities only required simple digital writes and analog reads from the microcontroller. This meant the priority was testing the far more complex GPS module, RockBLOCK communication modem, the serial camera, and the complex interplay between these systems.

The GPS module was tested from a high point with a view of the horizon until a lock was obtained and verified. The GPS monitor then stored its values for longitude, latitude, and altitude as part of a new normal report.

[illegible]

Figure 3: Successful normal reports as viewed from the RockBLOCK admin service

The Camera Reports were also successfully written to and read from the SD Card.

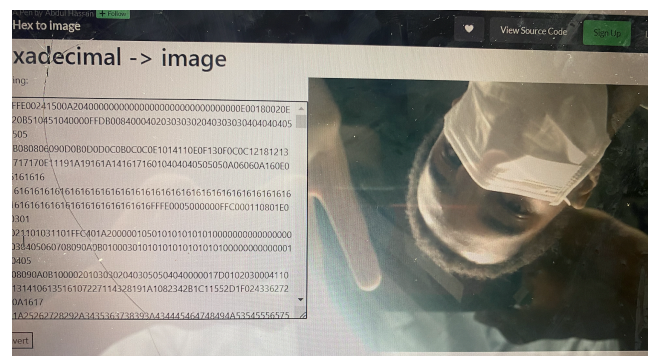
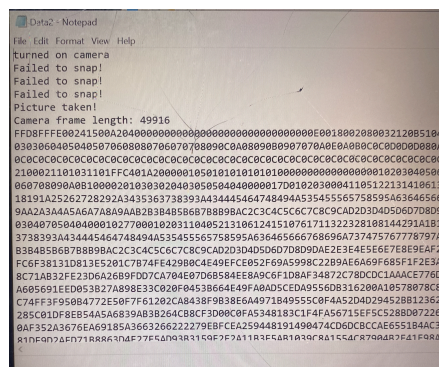


Figure 3: Capture of image as hexadecimal fragments. Figure 4 Conversion of fragments back into an image

The RockBLOCK control task again was able to send a series of camera reports. These reports are identified with their report flag of 42 (2a in hexadecimal). These reports include the image fragment ID and hexadecimal payload of a jpeg fragment.

Figure 6: Successful camera reports as viewed from the RockBLOCK admin service

02/Jul/2024 16:44:19	RockBLOCK 203136 (Balloon)	↓ MT	020000000003000030	9	1
02/Jul/2024 16:43:56	RockBLOCK 203136 (Balloon)	↓ MT	0100000493e000000000	10	1
02/Jul/2024 16:43:39	RockBLOCK 203136 (Balloon)	↓ MT	00000100000000000000	10	1
02/Jul/2024 16:43:12	RockBLOCK 203136 (Balloon)	↓ MT	00000000000000000000	10	1

Conclusion:

Going forward, the next steps would be front-end work. The development of a ground station for the mission would be used to facilitate human interaction with flight software. It would be a user interface that allows for the transmission of commands, the display of normal report packages, and compiling and displaying the fragmented images.